

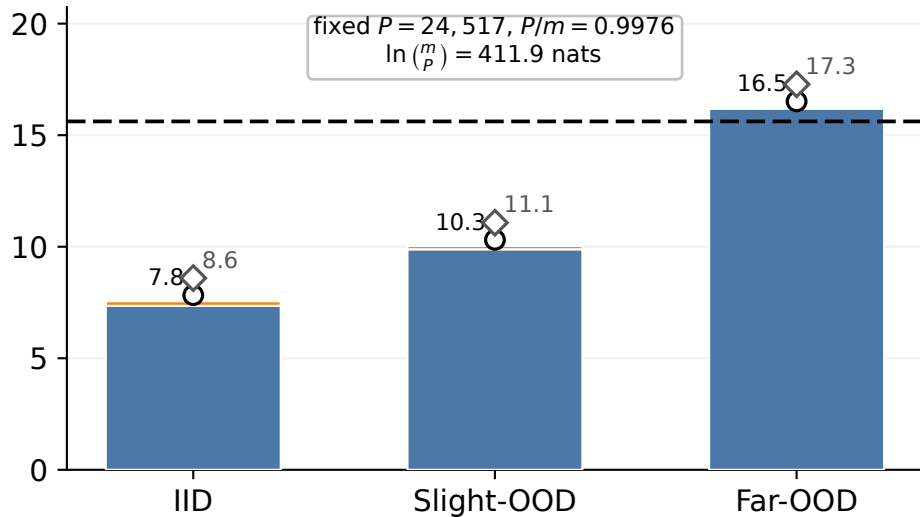
Certificate Decomposition under Synthetic Corruption (Top-k=64)

Empirical proxy risk $\hat{R}(h_G \cdot)$ Reconstruction gap $\hat{\epsilon}_{\text{loss}}$ Pool mismatch $\Delta\hat{\eta}$ $\circ$ U_{SAE} $\diamond$ $U_{\text{SAE}} + \text{Occam}$ $--$ R_{unif}

Certificate / loss contribution (bits)

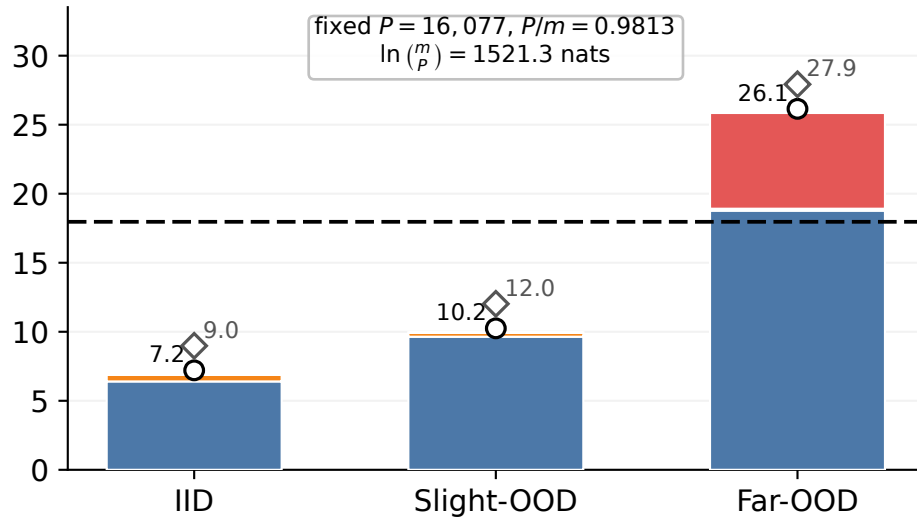
GPT-2 Small

fixed $P = 24,517$, $P/m = 0.9976$
 $\ln \binom{m}{p} = 411.9$ nats



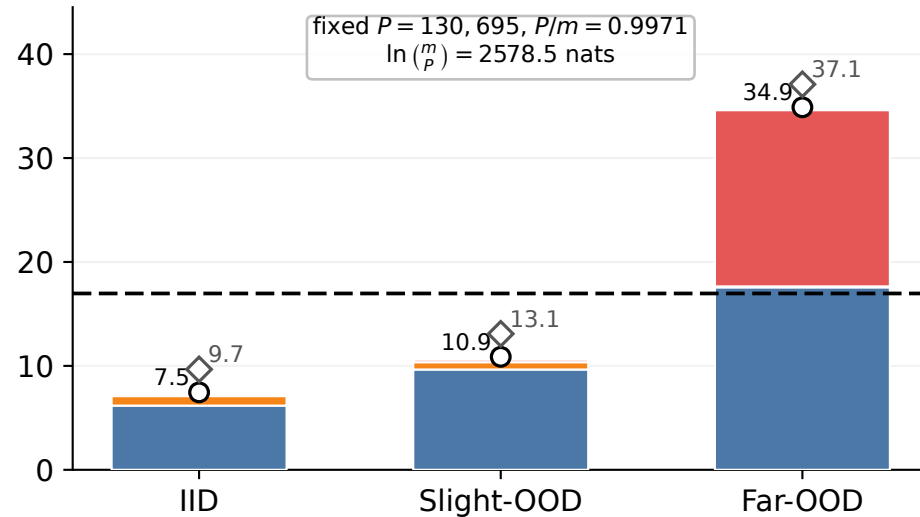
Gemma-2B

fixed $P = 16,077$, $P/m = 0.9813$
 $\ln \binom{m}{p} = 1521.3$ nats



Llama-3.1-8B

fixed $P = 130,695$, $P/m = 0.9971$
 $\ln \binom{m}{p} = 2578.5$ nats



Evaluation distribution